Part IV

Non-Standard Consequence Relations

13 Introduction

We shall now classify relations of logical consequence. Each principle or rule of logic enables us to distinguish between logical systems that satisfy it and those that do not.

We will examine some of these rules below, along with the systems that satisfy them and those that do not. The proposed subdivision is somewhat arbitrary, but it helps to organise the presentation of the rules.

Structural Principles | 14

Logical principles may be divided into two basic kinds: operational and structural principles.

Operational principles are the most well known as they involve logical operators, that is, expressions such as '$\neg$', '$\rightarrow$', '$\forall$', etc.

Structural principles, instead, specify which transformations of an inference preserve validity while abstracting entirely from the internal logical form of its formulae. They concern only the structural configuration of premises and conclusions.

In order to describe these principles, it is useful to introduce the following notation in which '$\mathbf{Cn}_\mathsf{S}$' will denote a function rather than a relation.

Definition 14.1 (Consequence set)**.**

$$\mathbf{Cn}_\mathsf{S}(\mathbf{X}) \overset{\text{DF}}{=} \{\mathtt{F} \in \mathbf{Fo}_\mathsf{S} : \langle \mathbf{X}, \{\mathtt{F}\}\rangle \in \mathbf{Cn}_\mathsf{S}\}.$$

That is, $\mathbf{Cn}_\mathsf{S}(\mathbf{X})$ is the set of all S-logical consequences of $\mathbf{X}$.

Let us consider the most well-known structural principles.

14.1 Extension aka Reflexivity

Say we have a set of formulae $\mathbf{X}$. In our notation, $\mathbf{Cn}_\mathsf{S}(\mathbf{X})$ would be the set of S-consequences of $\mathbf{X}$.

Intuitively, all formulae in $\mathbf{X}$ should also follow from $\mathbf{X}$. This property is called *extension* or *reflexivity*.

Definition 14.2 (Extension aka Reflexivity)**.**

$$\mathbf{Cn}_\mathsf{S} \text{ is extensive}^* \overset{\text{DF}}{=} \mathbf{X} \subset \mathbf{Cn}_\mathsf{S}(\mathbf{X}).$$

*: The term *reflexive logic* is also appropriate. However, there are some logical systems called 'non-reflexive' which are related to properties of the identity symbol.

Extension, or reflexivity, states that every set of formulae is included in its own consequence set. In other words, any set of premises entails at least itself.

There seems to be no motivation for extension to fail. However, consider a logical system in which we want the premises of a valid inference to provide reasons for the conclusion which are independent of the conclusion itself. Thus, no inference of the form $\mathtt{F}\ /\ \mathtt{F}$ should be valid under such system.

It is also noteworthy that Aristotle's syllogistic – as well as any adequate extension of it – is not extensive.

14.2 Idempotence

Consider a set of formulae $\mathbf{X}$ and its consequence set $\mathbf{Cn}_\mathsf{S}(\mathbf{X})$. Since $\mathbf{Cn}_\mathsf{S}(\mathbf{X})$ is also a set of formulae, we can also talk about $\mathbf{Cn}_\mathsf{S}(\mathbf{Cn}_\mathsf{S}(\mathbf{X}))$, that is, of the set of S-consequences of $\mathbf{X}$ of the set of S-consequences of $\mathbf{X}$.

Intuitively, if something follows from $\mathbf{X}$, it should also follow from $\mathbf{Cn}_\mathsf{S}(\mathbf{X})$. This property is called *idempotence*.

Definition 14.3 (Idempotence)**.**

$$\mathbf{Cn}_\mathsf{S} \text{ is idempotent} \overset{\text{DF}}{=} \mathbf{Cn}_\mathsf{S}(\mathbf{X}) = \mathbf{Cn}_\mathsf{S}(\mathbf{Cn}_\mathsf{S}(\mathbf{X})).$$

Idempotence guarantees that applying the consequence operation twice does not change the result. That is, once a set of consequences has been obtained, applying the consequence operation again will not produce any new conclusions, nor will it remove previous ones.

14.3 Increasingness and Monotonicity

Say we have two sets of formulae $\mathbf{X}$ and $\mathbf{Y}$ such that $\mathbf{X} \subseteq \mathbf{Y}$. It would be reasonable to think that, if something follows from $\mathbf{X}$, it should also follow from $\mathbf{Y}$, since every formula in $\mathbf{X}$ is also in $\mathbf{Y}$.

This property is called *increasingness*.

Definition 14.4 (Increasing)**.**

$$\mathbf{Cn}_\mathsf{S} \text{ is increasing} \overset{\text{DF}}{=} \mathbf{X} \subseteq \mathbf{Y} \Rightarrow \mathbf{Cn}_\mathsf{S}(\mathbf{X}) \subseteq \mathbf{Cn}_\mathsf{S}(\mathbf{Y}).$$

This reflects the idea that adding more premises may expand the set of deducible conclusions.

This property is often presented as a property of derivation systems, which is commonly known as *monotony* or *monotonicity*.

Definition 14.5 (Monotoncity)**.** $\mathbf{Cal}_\mathsf{S}$ is monotonous or monotonic $\overset{\text{DF}}{=} \mathtt{A}_1, \ldots, \mathtt{A}_n \ / \ \mathtt{F} \in \mathbf{Der}_\mathsf{S} \Rightarrow \mathtt{A}_1, \ldots, \mathtt{A}_n, \mathtt{B} \ / \ \mathtt{F} \in \mathbf{Der}_\mathsf{S}$.

Non-monotonic logics have been around for some decades now. Thus, it would make sense to do away with monotonicity when trying to capture defeasible reasoning; that is, cases in which, given a set of premises, we have a valid conclusion, but if we add more premises, the conclusion would no longer follow.

Consider, e.g., the following argument:

	All observed swans as of yesterday are white.	(♈_1)
∴	It is very plausible that all swans are white.	(♈_2)

This seems to be valid enough. However, let us augment a premise to get the following argument:

	All observed swans as of yesterday are white.	(♊_1)
	A black swan was observed today.	(♊_2)
∴	It is very plausible that all swans are white.	(♊_3)

The sole addition of (♊_2) has turned the valid argument (♈) into the non-valid argument (♊), which suggests a motivation to have non-monotonic or non-increasing logical systems.

14.4 Transitivity

Say we have three sets of formulae $\mathbf{X}, \mathbf{Y}, \mathbf{Z}$ such that $\mathbf{X} \subseteq \mathbf{Cn}_S(\mathbf{Y})$ and $\mathbf{Y} \subseteq \mathbf{Cn}_S(\mathbf{Z})$. It would be reasonable to conclude that $\mathbf{X} \subseteq \mathbf{Cn}_S(\mathbf{Z})$.

This property is called *transitivity*.

Definition 14.6 (Transitivity)**.**

$\mathbf{Cal}_S$ is transitive $\stackrel{\text{DF}}{=} \mathbf{X} \subseteq \mathbf{Cn}_S(\mathbf{Y}) \wedge\!\!\wedge\ \mathbf{Y} \subseteq \mathbf{Cn}_S(\mathbf{Z}) \Rightarrow \mathbf{X} \subseteq \mathbf{Cn}_S(\mathbf{Z})$.

Transitivity states that if a set of formulae **X** can be deduced from **Y**, and **Y** can be deduced from **Z**, then **X** can be deduced directly from **Z**.

This property is often presented as a property of derivation systems, which is commonly known as *cut*.

Definition 14.7 (Cut)**.** $\mathbf{Cal}_S$ features cut $\stackrel{\text{DF}}{=} \mathtt{X}_1, \ldots, \mathtt{X}_m \;/\; \mathtt{A} \in \mathbf{Der}_S \wedge\!\!\wedge\ \mathtt{A}, \mathtt{Y}_1, \ldots, \mathtt{Y}_n \;/\; \mathtt{B} \in \mathbf{Der}_S \Rightarrow \mathtt{X}_1, \ldots, \mathtt{X}_m, \mathtt{Y}_1, \ldots, \mathtt{Y}_n \;/\; \mathtt{B} \in \mathbf{Der}_S$.

Logics that do not feature the cut rule are usually known as *cut-free logics*, of which there are a number.

One reason to do away with transitivity is to investigate *default consequence* and not necessary consequence. This is the kind of consequence which is, so to speak, once inferential step away from its premises.

For instance, ‘Peter works at LMU’ is a default consequence of ‘Peter is a professor at LMU’; similarly, ‘Peter receives a salary from LMU’ is a default consequence of ‘Peter works at LMU’. However, ‘Peter receives a salary from LMU’ may not be regarded as a default consequence of ‘Peter is a professor at LMU’.

14.5 Further Properties

A consequence relation may be characterised in several ways as result of a combination of the properties defined above.

Definition 14.8 (Closure operation)**.** $\mathbf{Cn}_S$ is a closure operation $\overset{\text{DF}}{=}$ $\mathbf{Cn}_S$ is extensive, increasing, and idempotent.

Definition 14.9 (Tarskian consequence)**.** $\mathbf{Cn}_S$ is Tarskian $\overset{\text{DF}}{=}$ $\mathbf{Cn}_S$ is extensive, transitive, and monotonic.

Definition 14.10 (Substructural consequence)**.** $\mathbf{Cn}_S$ is substructural $\overset{\text{DF}}{=}$ $\mathbf{Cn}_S$ it does not feature at least one of the structural rules featured assumed by classical logic.

15 Operational Principles

Operational principles, unlike structural principles, are principles concerning operators.

It is relatively easy to write very large lists of interesting operational principles. Some of them correspond to basic inference rules of diverse logical systems, and others to important derived rules or tautologies in those systems.

We will classify those principles into ...

15.1 Principles about the Conditional

The principles of the first group concern the conditional. However, as some logical systems have more than one conditional, we will use the placeholder '☛' for greater abstraction.

The first such principle is known as *residuation*.

Definition 15.1 (Residuation)**.** S is ☛-residual $\overset{\text{DF}}{=}$

$$A_1, \ldots, A_n, B \ / \ C \in \mathbf{Cn}_S \Leftrightarrow A_1, \ldots, A_n \ / \ B \text{ ☛ } C \in \mathbf{Cn}_S.$$

Residuation is the property that allows us to convert the deduction of a formula C from a collection of premises $A_1, \ldots, A_n$ together with a formula B into the deduction of a compound formula B ☛ C from $A_1, \ldots, A_n$.

S0 and S1 are '$\longrightarrow$'-residual, since:

$$A_1, \ldots, A_n, B \ / \ C \in \mathbf{Cn}_{S0} \Leftrightarrow A_1, \ldots, A_n \ / \ B \longrightarrow C \in \mathbf{Cn}_{S0}.$$

A better-known name for this principle is *the deduction theorem*.

Task 15.1 (Residuation)**.**

- Determine whether the non-standard systems studied in this course are '$\longrightarrow$'-residual.
- Out of curiosity, determine whether S0 is '$\longleftrightarrow$'-residual or '$\vee$'-residual or '$\wedge$'-residual (or none).

Some principles strongly related to residuation principles may be called *detachment principles*.

Definition 15.2 (Detachment)**.** S is ☛-detaching $\overset{\text{DF}}{=}$ A, A ☛ $B \ / \ B \in \mathbf{Cn}_S$.

S0 and S1 are clearly '$\longrightarrow$'-detaching, since:

$$\mathtt{A}, \mathtt{A} \longrightarrow \mathtt{B} \;/\; \mathtt{B} \in \mathbf{Cn}_{\mathsf{S}}.$$

This specific detachment principle is known as or *modus ponendo ponens* (MPP).

Task 15.2 (Detachment)**.**

- ▶ Determine whether the non-standard systems studied in this course are '$\longrightarrow$'-detaching.
- ▶ Out of curiosity, determine whether S0 is '$\leftrightarrow$'-detaching or '$\vee$'-detaching or '$\wedge$'-detaching (or none).

15.2 Detonation Principles

The principles of this group concern a phenomenon of most logical system which may be called *embracingness* or *detonation*. These principles usually concern the negation, for which we will use the placeholder '$\odot$'.

The following are some denotation principles.

Definition 15.3 (Explosion)**.** S is $\odot$-explosive $\overset{\text{DF}}{=}$ $\mathtt{A}_1, \ldots, \mathtt{A}_n \;/\; \mathtt{B} \in \mathbf{Cn}_{\mathsf{S}} \mathbin{\wedge\!\!\!\wedge} \mathtt{A}_1, \ldots, \mathtt{A}_n \;/\; \neg\mathtt{B} \in \mathbf{Cn}_{\mathsf{S}} \Longrightarrow \mathtt{A}_1, \ldots, \mathtt{A}_n \;/\; \mathtt{C} \in \mathbf{Cn}_{\mathsf{S}}$.

S0 and S1 are clearly '$\neg$'-explosive. This specific explosion principle is known as *ex contradictione sequitur quodlibet* (ECQ).

Task 15.3 (Explosion)**.**

- ▶ Determine whether the non-standard systems studied in this course are '$\neg$'-explosive.
- ▶ Given a system $\mathsf{C0}_n$ (with $1 < n < \omega$), determine whether it is: '$\overset{(n-1)}{\neg}$'-explosive, or '$\overset{(n)}{\neg}$'-explosive, or '$\overset{(n+1)}{\neg}$'-explosive.
- ▶ Out of curiosity, determine whether S0 is '$\neg\neg$'-explosive.

A system which fails to be '$\neg$'-explosive is '$\neg$'-paraconsistent.

Definition 15.4 (Paraconsistency)**.** S is $\odot$-paraconsistent $\overset{\text{DF}}{=}$ S is not $\odot$-explosive.

In paraconsistent systems it is not possible to derive anything whatsoever from two opposing premises $\mathtt{A}$ and $\neg\mathtt{A}$.

Task 15.4 (Paraconsistency).

- Determine whether the non-standard systems studied in this course are '$\neg$'-paraconsistent.
- Given a system $\mathtt{CO}_n$ (for $n > 1$), determine whether it is: '$\overset{(n-1)}{\neg}$'-paraconsistent, or '$\overset{(n)}{\neg}$'-paraconsistent, or '$\overset{(n+1)}{\neg}$'-paraconsistent.
- Out of curiosity, determine whether $\mathtt{S0}$ is '$\neg\neg$'-paraconsistent.

We can define a strong notion of paraconsistency:

Definition 15.5 (Strong paraconsistency). $\mathtt{S}$ is strongly $\odot$-paraconsistent $\overset{\text{DF}}{=} \forall \mathtt{A}_1 \ldots \forall \mathtt{A}_n \, \forall \mathtt{B} \, \forall \mathtt{C} \, (\mathtt{A}_1, \ldots, \mathtt{A}_n \;/\; \mathtt{B} \in \mathbf{Cn}_{\mathtt{S}} \;⩕\; \mathtt{A}_1, \ldots, \mathtt{A}_n \;/\; \odot\mathtt{B} \in \mathbf{Cn}_{\mathtt{S}} \;⩕\; \mathtt{A}_1, \ldots, \mathtt{A}_n \;/\; \mathtt{C} \notin \mathbf{Cn}_{\mathtt{S}})$.

Moreover, we can define a more abstract version of the idea of explosion, which could well be considered a structural-type rule.

Definition 15.6 (Detonation). $\mathtt{S}$ is detonating or embracing* $\overset{\text{DF}}{=} \exists \mathbf{M} \, (\mathbf{M} \subset \mathbf{Fo}_{\mathtt{S}} \;⩕\; \mathbf{Cn}_{\mathtt{S}}(\mathbf{M}) = \mathbf{Fo}_{\mathtt{S}})$.

*: The term 'detonation' comes from Scotch and Jennings (1989), who argued that a logic without this property would imply 'a completely new conception of inference, which violates the classical theory of structural rules'. The term 'embracing' comes from Karl Popper (1943), who argued that, even if we rejected $\neg$-explosion, logical systems still need to be embracing in the sense mentioned above.

Note that even a strongly paraconsistent logic can be detonating, as there can be several sets of formulae able to generate arbitrary conclusions.

In this regard, it is worth mentioning a type of system that, although useless, is often mentioned as a limiting case of a logical system.

Definition 15.7 (Trivial logic). $\mathtt{S}$ is **trivial** $\overset{\text{DF}}{=} \forall \mathbf{M} \, \mathbf{Cn}_{\mathtt{S}}(\mathbf{M}) = \mathbf{Fo}_{\mathtt{S}}$.

It is worth mention here the other systems have attempted to avoid explosion by introducing a notion of relevance. In this case, it would not be sufficient for the premises to entail the conclusion according to the standard rules, but the former should also be relevant to the latter.

The very notion of relevance is nevertheless difficult to explicate. A suggested principle has been that of *variable sharing* between premises and conclusion. Logical systems in which this relevance from premises to conclusion is pursued are known as *relevance logics* or *relevant logics*.

15.3 Classical or Aristotelian principles

There are some principles typically associated to Aristotle. However, it is debatable to what extent he defended these principles, or whether he regarded them as logical ones.

Definition 15.8 (Non-contradiction)**.** $\mathtt{S}$ is non-contradictory $\overset{\text{DF}}{=} \forall \mathbf{M}\, \ulcorner\neg(\mathtt{A} \wedge \neg\mathtt{A})\urcorner \in \mathbf{Cn}_{\mathtt{S}}(\mathbf{M})$.

The denomination 'paraconsistent logic' was at first used to denominate systems which aren't non-contradictory. However, it was later used to denominate systems which are non-explosive.

Definition 15.9 (Excluded middle)**.** $\mathtt{S}$ satisfies the *principle of excluded middle* or *tertium non datur* $\overset{\text{DF}}{=} \forall \mathbf{M}\, \ulcorner(\mathtt{A} \vee \neg\mathtt{A})\urcorner \in \mathbf{Cn}_{\mathtt{S}}(\mathbf{M})$.

A system which does not satisfy this principle is called *paracomplete*. Among them are many-valued logics (with more than two truth values), fuzzy logics, intuitionistic logics, among many others.

Further Aristotelian principles are those concerning connexivity and identity, i.e., for all $\mathtt{t}$, $\mathtt{t} = \mathtt{t}$.